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Wireless Network Security Assessment

Comprehensive Vulnerability & Rogue AP Audit

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Prepared For: Executive Management & IT Security Teams

Assessment Scope: Internal Wireless Infrastructure

STRICTLY CONFIDENTIAL

1. Executive Summary

A comprehensive wireless network security audit was conducted to evaluate the current security posture of the organizational Wi-Fi infrastructure. The objective of this assessment was to identify potential vulnerabilities, detect unauthorized rogue access points, and analyze network traffic for exposure of sensitive data.

Key Finding: The overall security posture is moderately secure. No critically vulnerable WEP networks were identified. However, several networks operate in a mixed WPA/WPA2 mode which introduces downgrade attack vectors. Connected devices are authenticated properly, and no rogue Access Points (Evil Twins) were detected during the sweep.

2. Assessment Scope & Methodology

The security assessment was performed utilizing industry-standard Blue Team methodologies, adhering to the following phases:

- **Reconnaissance:** Passive RF scanning using NetSpot to enumerate broadcasted SSIDs, BSSIDs, and cryptographic protocols.
- **Device Profiling:** ARP/ICMP sweeps utilizing command-line network mapping to detect active endpoints on the internal subnet.
- **Traffic Interception:** Promiscuous packet capture via Wireshark to analyze HTTP/DNS payloads and verify encryption standards.

3. Wireless Network Enumeration

A wide-area RF scan was executed to identify all surrounding wireless networks and evaluate their encryption standards. A total of 16 networks were successfully enumerated.

3.1 Encryption Analysis

A review of the cryptographic implementations revealed the following distribution:

Encryption Standard	Count	Risk Level	Assessment Note
WEP / Open	0	None	No critically vulnerable networks detected.
WPA2-Personal	9	Low	Standard encryption; vulnerable only to offline dictionary attacks if passwords are weak.
WPA/WPA2 Mixed	7	Medium	Allows fallback to older WPA (TKIP) standards, increasing susceptibility to interception.

Observation: The primary testing network ("Airtel_AMAAR 05") is currently utilizing WPA/WPA2 mixed mode. While this provides legacy device compatibility, it violates modern enterprise security compliance standards.

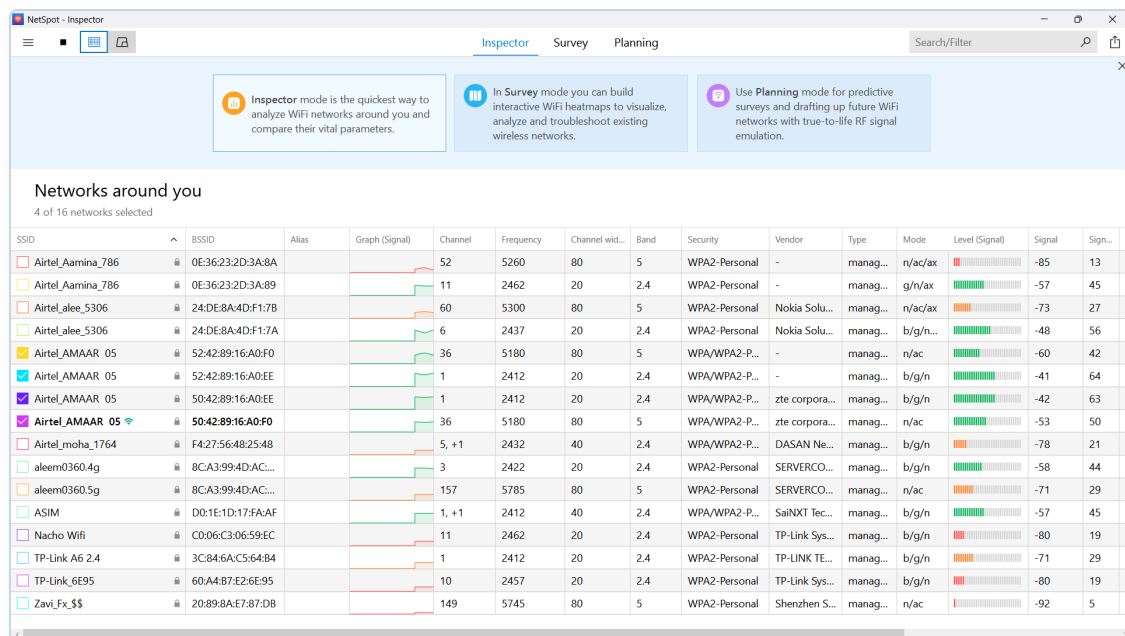


Figure 1.0: NetSpot RF Reconnaissance detailing SSIDs, BSSIDs, and security configurations.

4. Endpoint Discovery & Rogue AP Detection

To detect unauthorized presence and verify the integrity of the internal network, an ARP sweep was executed against the primary subnet (192.168.1.0/24). This process maps active physical addresses (MAC) against assigned IPs.

4.1 Internal Topology Audit

The network mapping identified the default gateway and several authenticated clients. No unexpected, unauthorized, or disguised hardware signatures (such as rogue Raspberry Pi drops or unauthorized repeaters) were detected in the routing tables.

```
Command Prompt
Microsoft Windows [Version 10.0.26200.8737]
(c) Microsoft Corporation. All rights reserved.

C:\Users\abdxl>arp -a

Interface: 192.168.1.6 --- 0x12
Internet Address      Physical Address      Type
192.168.1.1           50-42-89-16-a0-ee     dynamic
192.168.1.4           2e-dc-7a-6d-62-1a     dynamic
192.168.1.5           b0-f2-f6-bc-1b-1c     dynamic
192.168.1.11          6a-69-85-0a-e4-e2     dynamic
192.168.1.255         ff-ff-ff-ff-ff-ff     static
224.0.0.22            01-00-5e-00-00-16     static
224.0.0.251           01-00-5e-00-00-fb     static
224.0.0.252           01-00-5e-00-00-fc     static
239.255.255.250       01-00-5e-7f-ff-fa     static
255.255.255.255       ff-ff-ff-ff-ff-ff     static
```

Figure 2.0: Command-line ARP sweep mapping the local topology and active endpoint signatures.

5. Packet Capture & Traffic Analysis

Deep Packet Inspection (DPI) was performed to observe the transmission of data frames between local clients and the Access Point. Wireshark was utilized to intercept and analyze standard network telemetry.

5.1 Traffic Integrity

The captured traffic predominantly consisted of standard TCP handshakes, ARP broadcasts, and secure TLSv1.2/TLSv1.3 encrypted payloads. To demonstrate the vulnerability of unencrypted transmission, a test payload was intentionally generated over standard HTTP.

Observation: Modern traffic is heavily encrypted via HTTPS/TLS, mitigating the risk of packet sniffing on the local subnet. However, DNS queries remain largely unencrypted, revealing endpoint browsing behaviors.

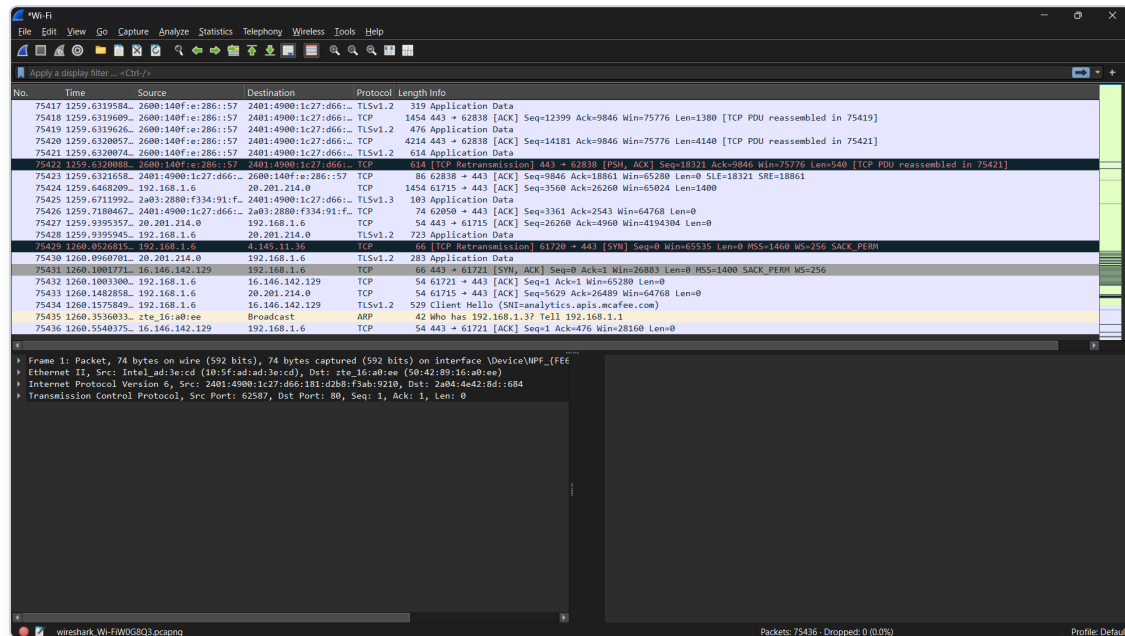


Figure 3.0: Wireshark packet capture displaying TCP retransmissions, TLS handshakes, and ARP requests.

6. Strategic Recommendations & Remediation

Based on the findings of this security audit, the following remediations are recommended to elevate the organization's defensive posture to enterprise standards:

- **Deprecate Legacy Protocols (High Priority):** Immediately disable WPA/WPA2 Mixed Mode on all corporate Access Points. Enforce strict WPA2-AES or upgrade to WPA3-SAE to prevent downgrade attacks.
- **Implement Endpoint Isolation (Medium Priority):** Configure AP isolation (Client Isolation) on guest networks to prevent lateral movement and packet sniffing between connected devices.
- **Deploy Continuous Monitoring (Medium Priority):** Utilize Intrusion Detection Systems (IDS) to automatically flag the sudden appearance of Evil Twin SSIDs or unauthorized MAC address cloning.
- **Enforce DNS over HTTPS (DoH) (Low Priority):** Configure network endpoints to utilize DoH to prevent passive interception of browsing habits and DNS spoofing attacks.

-- End of Assessment Report --
